To what extent has private investment in human capital complemented government initiatives in increasing the supply of required workforce for the growing fintech industry in Singapore, since 2020?

Bella Chien

November 13th, 2023

Table of Contents

1 Introduction **2**
2 Methodology **2**
3 The Complementary Relationship of the Private and Public Sector **3**
4 Fintech Industry of Singapore **6**
4.1 Background of the Fintech Industry 6
4.2 The Labor Shortage 7
4.2.1 The Quantity of Labor 7
4.2.2 The Quality of Labor 8
5 Wage Factors **10**
5.1 Obstacles in Wage Determination 10
5.2 Initiatives and Investments of the Private and Public Sector 11
5.2.1 Government Initiative: Wage Subsidies 11
5.2.2 Private Sector Response to Wage Subsidies 12
5.3 Evaluation Using Wage Elasticity of Labor Demand and Supply 13
6 Non-wage Factors **13**
6.1 Impact of COVID-19 on Singapore's Fintech Industry 13
6.2 Non-monetary Employee Benefits 15
6.3 Upskilling the Workforce 15
6.4 Reskilling the Workforce 17
7 Alternatives **17**
7.1 Artificial Intelligence and Other Capital Substitution to Increase Productivity 17
7.2 Foreign Workers 19
8 Conclusion **20**
9 References **21**
10 Appendices **31**
Appendix A 31

1 Introduction

Amidst the pandemic, the market of fintech experienced significant growth with the digitalization of business processes and daily life (Fu & Mishra, 2021). Singapore, despite its small population of 6 million people, quickly established itself as a leader in the industry (Worldometer, 2023). From 2020, the value of fintech investments in Singapore increased 78.3% to a total of USD4.1B in 2022, and currently, Singapore's fintech market is valued at USD7.8B (Statista Research Department, 2023a). Moreover, Singapore's fintech industry is shown to be positively correlated with multiple economic indicators including GDP, GDP per capita, R&D intensity as well as labor productivity (Singapore Business Review, 2023b).

Despite taking the place of "Asia's top fintech hub", Singapore suffers from a severe labor shortage in tech-related talent (Singapore Business Review, 2023a; Yahya, 2022). The government and private firms acknowledge this as an obstacle that must be overcome to sustain the desired growth of the market. As a result, they have pursued multiple endeavors including training, immigration policies, digitization and more to bridge the skill gap while reducing their reliance on foreign labor. Therefore, this essay aims to evaluate the case study of the labor shortage in Singapore's fintech industry to answer the question, **"To what extent private investment in human capital has complemented government initiatives in increasing the supply of required workforce for the growing fintech industry in Singapore, since 2020?"**

2 Methodology

Statista.com is used for a fundamental understanding of the fintech industry and Singapore's labor market. News and opinion articles, primarily from Fintech News, Reuters, Business Times and Linkedin, are also used. To affirm their credibility, facts and statistics are cross-verified with other similar sources while the normative statements are authenticated through academic publications on JSTOR and academia.

To understand the economic theories behind the dynamics of private investment and government initiatives, the essay applied those exemplified in Germany's Vocational Education and Training (VET) system, a successful example that has inspired many public and private sector collaborations in human capital in the present.

For more nuanced research on the industry, annual reports produced by Singaporean government organizations, such as the Monetary Authority of Singapore (MAS) and the Singapore Fintech Association (SFA), in collaboration with professional service firms including PwC and Accenture are used. Their website also provides a comprehensive list of governmental support to promote its fintech industry.

3 The Complementary Relationship of the Private and Public Sector

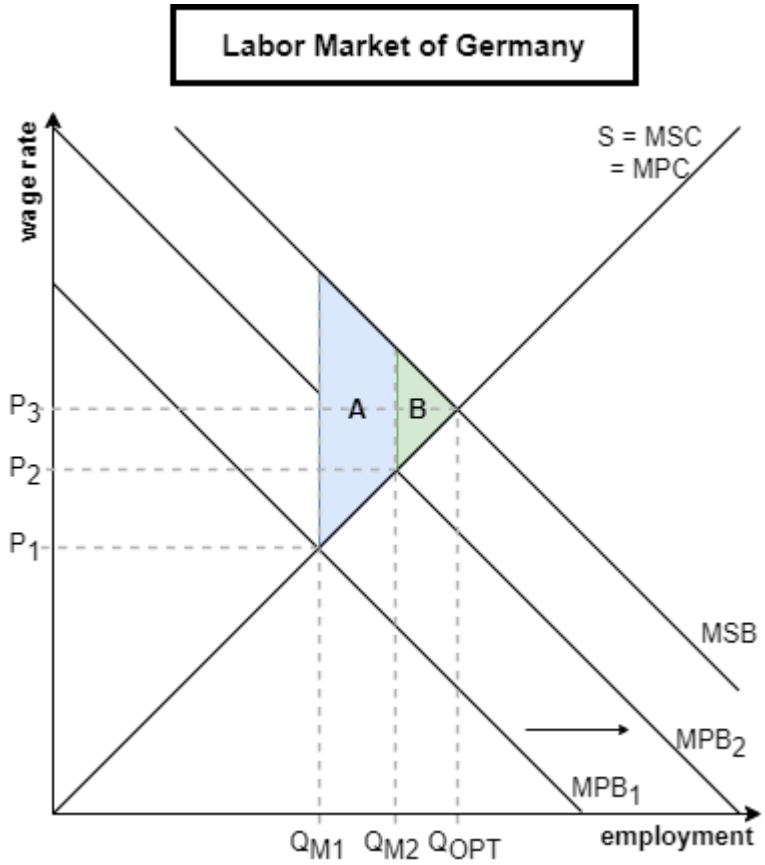


Figure 1. Correcting positive consumption externality in the form of youth unemployment

In the early 20th century, Germany faced high youth unemployment rates (Harvey, 1993). As observed in *fig. 1*, when firms employ fresh graduates, it creates a social spillover benefit in the form of improvement in labor productivity, social well-being of young people and overall economic stability , causing marginal social benefit (MSB) to be greater than marginal private benefit (MPB) in the labor market (Deloitte, 2015). However, firms were reluctant to hire inexperienced youth and provide training as it increased costs and reduced productivity. Hence, market equilibrium (Q_{M1}) is less than the social optimum (Q_{OPT}), indicating an underallocation of resources and a welfare loss of area A+B.

To decrease the externality, the German government opened up numerous public vocational schools, shifting MPB_1 to MPB_2 which reduced the area of welfare loss from A+B to B as it improved candidates' employability (Pinnow, 2019).

As technology advanced, the demanded skills became more specialized but the curriculum designed by the government lacked practical experience and did not meet the needs of the private sector. In 1969, the government enacted the VET system which allowed the input of employer representatives and trade unions in coordinating with the government to design a curriculum (GOVET, n.d.). The system integrated in-school education with work-based training which provided a smooth transition between school and work (Federal Institute for Vocational Education and Training, n.d.).

This significantly and sustainably reduces youth unemployment rates. Even during COVID-19 where global youth unemployment rates rose to 18.4%, Germany's figure remained low at 7.13% (O'Neill, 2023a; O'Neill, 2023b). This shifts MPB_2 even closer to MSB, minimizing welfare loss. Beyond a decrease in the cost of on-the-job training, it brought macroeconomic spillover benefits including an improvement in initial productivity,

and quality of skills, contributing to development in production and service delivery (Haasler, 2020). This shifts $LRAS_1$ (long-run aggregate supply) rightward, to $LRAS_2$, as seen in *fig. 2*, indicating a greater capacity for real output when all resources are fully utilized.

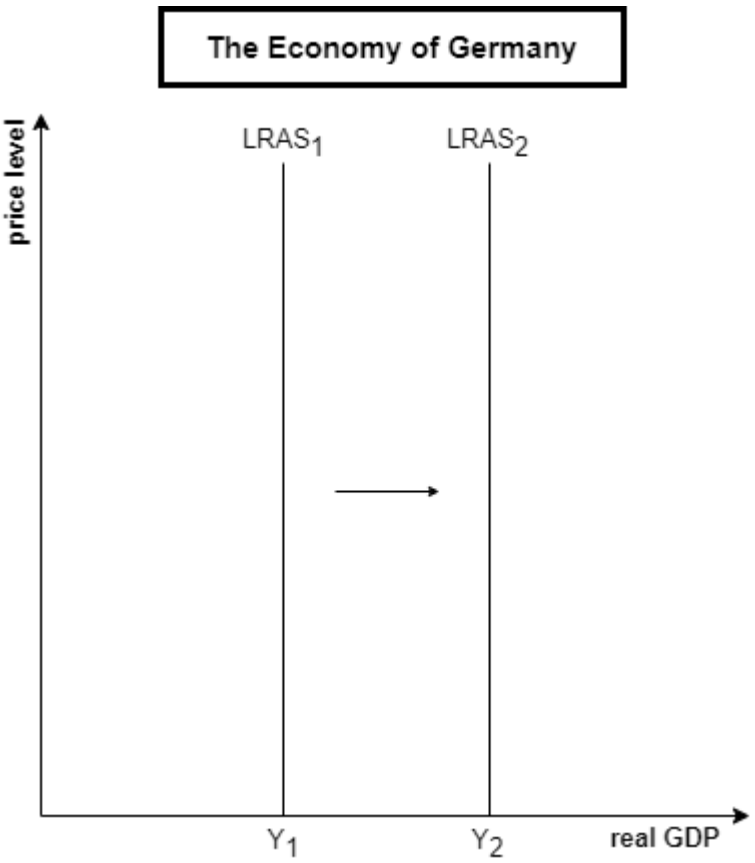


Figure 2. LRAS shifts rightward due to increase in labor productivity from the VET system

This example offers theoretical insights into addressing the government's oversight of private sector needs and producers' reluctance to provide training due to high costs, which can be applied to Singapore's fintech. The benefits and effectiveness of such endeavors can also be predicted.

4 Fintech Industry of Singapore

4.1 Background of the Fintech Industry

Singapore boasts one of the most developed financial industries globally with the industry contributing to 13% of the country's GDP, ranking it third, behind the business services and

retail industry (Department of Statistics Singapore, 2023; IndSights Research, 2023). In recent years, the government has directed significant resources towards technological advancement, investing in R&D and manpower development, resulting in a 8.6% GDP growth by the IT sector in 2022, making it the third highest contributor to the country's economic expansion (Kumar & Siddique, 2010; Statista Research Department, 2023c).

Fintech is a growing strategic industry, branching from finance and technology. Its investments accounted for 1.11% of its GDP in 2022, compared to an average of 0.25% worldwide, which indicates the significance of Singapore's fintech industry relative to GDP (Singapore Business Review, 2023b).

Building on the success of the FSTI scheme (Financial Sector Technology and Innovation) that awarded over SDG300M to more than 1500 projects, the government further committed SDG250M in the follow-up scheme, FSTI 2.0 in August 2020 (Fintech News Singapore, 2020). Moreover, in response to the pandemic, the MAS announced a SDG125M support package in April 2020 for all financial institutions (Monetary Authority of Singapore, 2020).

4.2 The Labor Shortage

Across various industries around the world, firms have found it increasingly difficult to secure the talent they require with global talent shortages reaching 77% as of 2023 (ManpowerGroup, 2022). Singapore has suffered slightly more with 84% of employers struggling to find a suitable hire in 2022 (ManpowerGroup, 2023).

4.2.1 The Quantity of Labor

The exact number of the shortage has not been explicitly stated. However, following the calculations in *Appendix A*, approximations can be made by looking at the job vacancies of the IT and fintech sector along with the number of IT graduates. As observed in *fig. 3*, the

fintech sector is likely to have experienced a local shortage of 842 jobs in 2021. In the following year, both supply of IT graduates and demand for fintech services increased, but demand exhibited a proportionately greater rise. The prominent rightward shift in the demand curve from D_{2021} to D_{2022} and the relatively subtle shift in supply from S_{2021} to S_{2022} signals greater competition for skilled tech labor. This results in an exacerbated shortage and an upward pressure on wages which justifies the increase in wages from SDG3958 to SDG4750. Despite this, it can be observed that the wages offered in fintech are still lower than the equilibrium wage, and thus, making it less competitive than other firms seeking tech talent.

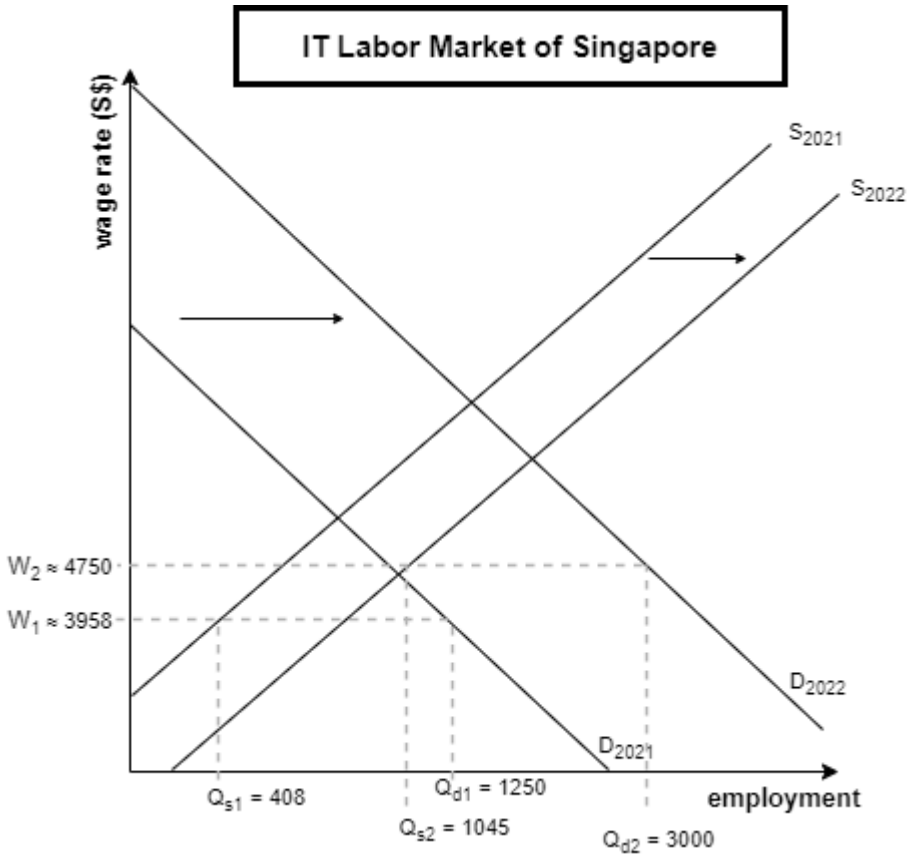


Figure 3. Comparison of the shortage in the IT labor market of Singapore between 2021 and 2022

However, it must be noted that this is an optimistic estimation. While the supply could be higher due to mid-career switches, this is likely to play a marginal impact (further elaborated in section 6). This is also outweighed by a multitude of other factors decreasing the estimated supply, including graduates leaving the country or pursuing higher education,

the probability of individuals choosing a job in fintech compared to another industry with a better job offer or the turnover rate to a job unrelated to fintech. Moreover, the birth rates rapidly decrease with 8.336 births per 1000 people as of 2023, a nearly 2% decline from the birth rate in 2020 (Macrotrends, 2023; Cheng, 2023). Although in the short-term it may imply an increase in the supply of women in the workforce, without effective government policies to counteract the 0.65% population growth rate, it could exacerbate the shortage in the long-term (Macrotrends, 2023). Compounding this, the increasing number of retirees is a prevailing problem: in 2020, 1 in 6 citizens were aged 65 and above, but currently, this number has risen to 1 in 5 (Lee, 2023). Therefore, it is likely that the shortage problem is even more severe than what is calculated.

4.2.2 The Quality of Labor

Jobs related to software and application development have always been and continue to be in high demand (SFA, Banking and Financial Services Union, & PwC, 2020; SFA & Accenture, 2022). However, within this category of jobs, the skills in demand are also changing. In 2022, skills related to applied intelligence, blockchain and cyber security have grown in demand while other skills, like database management, are declining (SFA & Accenture, 2022).

As the sector matures, many fintech businesses are at a stage where they are looking to expand or accelerate revenue growth. Hence, in 2022, there was a surge in demand for jobs related to business development including sales and business development (Indeed Editorial Team, 2023; SFA & Accenture, 2022; Tong, 2023). It is said that they help “companies do more with less” suggesting that these roles may help improve productivity by optimizing opportunities and sales prospects (Chandra, 2018).

Despite both categories of jobs being in demand, the level of urgency to resolve the gap greatly differs. The number of post-secondary graduates of business studies is approximately 50% higher than the number of IT graduates every year (Ministry of

Education, 2020; Ministry of Education, 2021; Ministry of Education, 2022). Additionally, business development roles have lower barriers of entry and better transferability, so firms were willing to hire candidates with no experience, compared to roles with skill requirements like AI and cybersecurity where firms preferred candidates with at least one year's work experience (PwC, Banking and Financial Services Union, & SFA, Banking and Financial Services Union, & PwC, 2020). As there are existing mechanisms in place, less effort needs to be invested in closing the shortage in business development compared to the technology aspect of fintech.

5 Wage Factors

5.1 Obstacles in Wage Determination

The industry must contend with the salary inflation which means candidates expect higher pay than offered. Indeed, 67% of industry leaders have claimed this to be one of the drivers of the fintech talent gap (SFA & Accenture, 2022).

Firms in the industry are not only competing against each other, but they are also forced to compete with other sectors for tech talent. The median monthly income in Singapore is SDG5070 per month as of 2022 (ExplainThis, 2023). Well-known software manufacturers pay around SDG6300~9800 per month for a job as a junior engineer whereas smaller software companies offer around SDG4000~9500 (ExplainThis, 2023). The latter has a lower and wider range of salaries, suggesting possible job insecurity, and when comparing the jobs solely by monetary rewards, large software firms are usually at an advantage (SFA, Banking and Financial Services Union, & PwC, 2020).

Tech jobs also have large disparities in salary based on seniority compared to other jobs. While other case studies found that this increases productivity and reduces turnover costs as it attracts workers who plan to stay at the company in the long-term (Bayo-Moriones, Galdón-Sánchez, & Güell, 2004), it may not be applicable in this context. Firms are already

struggling to obtain talent, if they actively eliminate candidates with short tenure, their pool of labor supply will significantly reduce. However, there is no substantial evidence confirming the impact of the large salary ranges in this context.

5.2 Initiatives and Investments of the Private and Public Sector

5.2.1 Government Initiative: Wage Subsidies

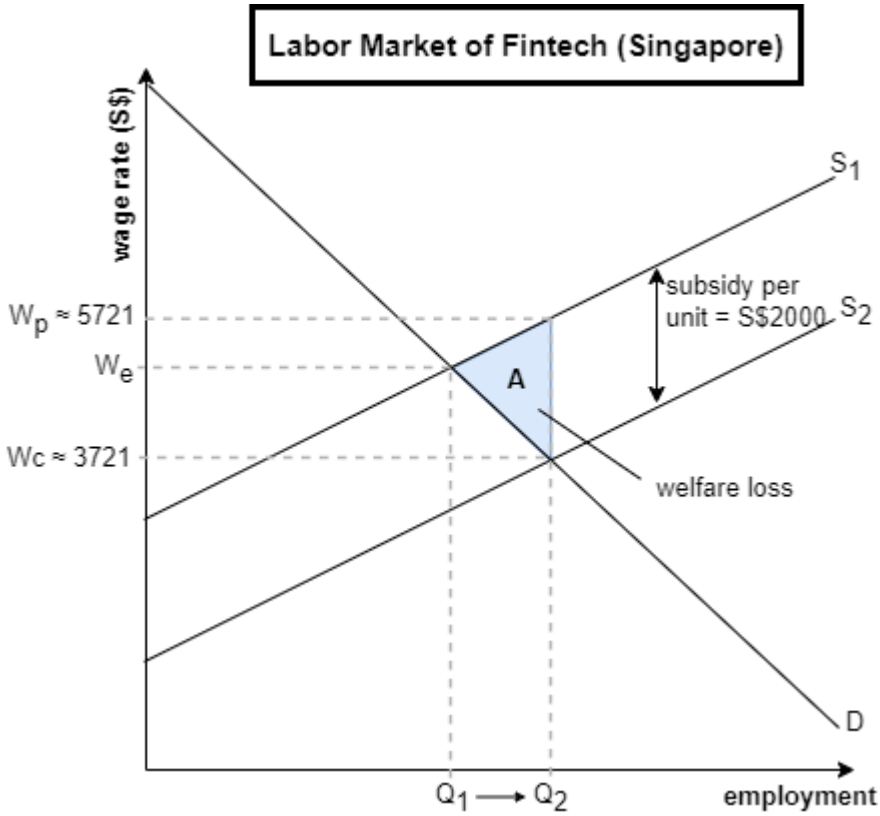


Figure 4. Effect of the Fintech Solidarity Grant's wage subsidy on the labor market of fintech (Singapore)

Answering to these difficulties, MAS launched a SDG6M Fintech Solidarity Grant in which there are various wage support programs, including up to SDG2000 per month per local staff as indicated by the vertical difference between S_1 and S_2 as illustrated in *fig. 4*.

However, even in 2023, junior software engineers in the finance sector received a median monthly salary of SDG5721, much lower than the SDG6829 median in the IT sector (The Strait Times, 2023). Hence, it can be assumed that prior to the subsidy, fintech firms

only paid at W_e, seeing as salaries rose approximately 5% to 10% in 2021 (Lim, 2022) and then another 22% to 32% in 2022 (INSIGHT, 2022).

5.2.2 Private Sector Response to Wage Subsidies

There are a few explanations for the limited effect of the government initiatives.

The first problem is that surveys show that 69% of FSTI applicants saw an increase in headcount yet it is not leveraged to their fullest potential as 80% of firms do not apply for it, mostly due to the lack of awareness (SFA & Accenture, 2022).

The second problem is that the private sector does not view higher pay as the tipping point to closing the talent gap. “Higher pay” has continuously been ranked as the top driver of the talent gap (SFA, Banking and Financial Services Union, & PwC, 2020; SFA & Accenture, 2022). However, “offer[ing] higher salary packages” was believed to be a relatively ineffective solution to bridging the gap in 2020 (SFA, Banking and Financial Services Union, & PwC, 2020). This means the private sector leaders do not see wages as a sustainable method in reducing the shortage and therefore do not place emphasis on it.

Though not explicitly stated, it can be inferred that the impracticality of this solution lies in the fact that pay rises would significantly inflate business costs which is difficult for smaller fintech firms that often operate on a lean business model. This is attributed to the fact that only a 25% to 30% pay raise from their previous job is considered competitive enough for candidates to switch jobs (Lim, 2022). Hence, candidates tend to expect higher pay.

Fortunately, the subsidy still provides benefits as it reduces labor costs to Wc in *fig. 4* which enables firms to invest their funds in other areas. Indeed, in 2020, only 58.8% of firms dedicated large funds toward manpower whereas 70.6% directed it towards the “implementation of digital solutions” (SFA, Banking and Financial Services Union, & PwC,

2020). While this may not increase the supply of labor, this form of factor substitution may improve productivity and therefore reduce the demand for labor.

5.3 Evaluation Using Wage Elasticity of Labor Demand and Supply

Salaries of software engineers have increased by an average of 7.6% in 2022 (Chiang, 2023). Yet, in 2022, JobStreet Singapore (a prominent online job portal) reported a year-on-year growth of 63% in job advertisements for tech-related positions in the same year (Digital News Asia, 2023). While these findings are not exclusive to software engineers in fintech, given that software engineers are one of the most in-demand tech professionals of 2023, it can be inferred that their demand is likely to be of a similar or even higher percentage.

This derives an elastic labor demand of 8.30, yet despite the increased salaries, the labor demand increases significantly, implying that the benefits of hiring tech talent outweighs the higher cost of labor, highlighting the significance of labor to these firms.

However, only a 2.34% increase was seen in the number of IT graduates in 2022, producing an inelastic labor supply of approximately 0.31 (Ministry of Education, 2022). The fintech sector can view this optimistically as it suggests that wage factors are not what attracts individuals to work in the sector. This justifies the argument of fintech firms in which increasing wages may not be a solution.

6 Non-wage Factors

6.1 Impact of COVID-19 on Singapore's Fintech Industry

COVID-19 catalyzed the adoption of electronic payment alternatives and various other fintech services as 66% of fintech firms saw an increase in demand in their business (SFA, Banking and Financial Services Union, & PwC, 2020). It was identified in 2020 that digital banking, one of the sectors that have seen significant growth in demand due to social distancing, is believed to boost the supply of local talent. This is because it is perceived as the

intersection of banking and fintech, so the type of work seems more familiar and less daunting, reducing the perceived job insecurity mentioned in section 5 (SFA, Banking and Financial Services Union, & PwC, 2020). As a result, there has been growing interest in the sector.

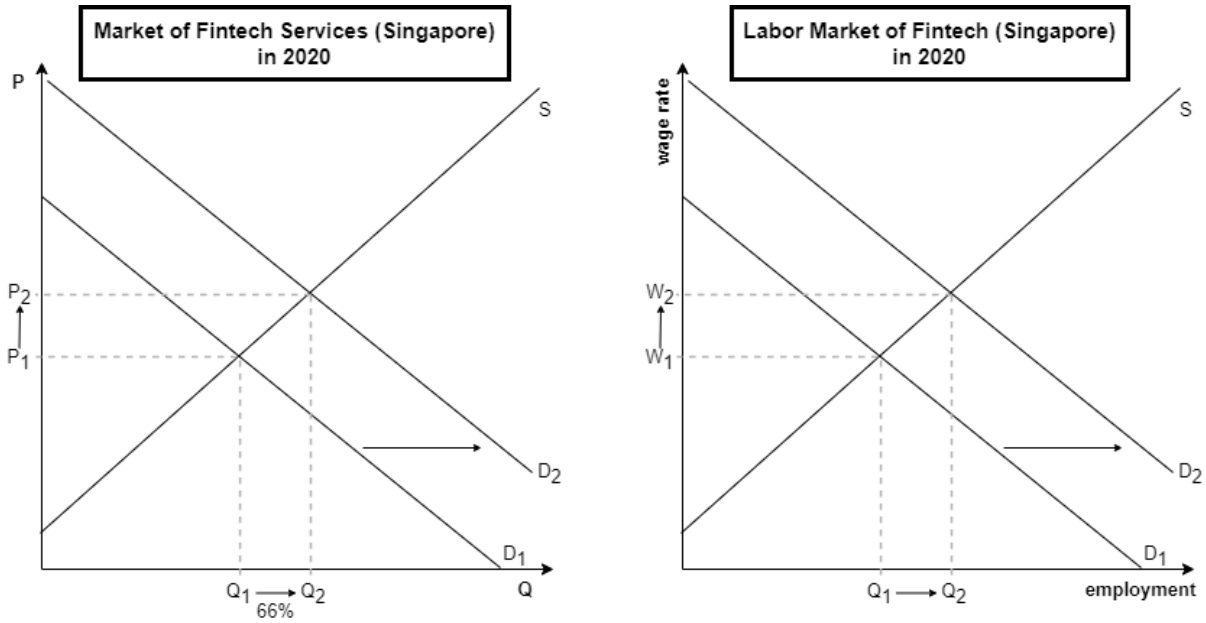


Figure 5. Effect of increase in demand of fintech services during COVID-19 on the derived demand of labor

In 2020, 68% of businesses in fintech believe that COVID-19 positively impacted the industry (SFA, Banking and Financial Services Union, & PwC, 2020). This had a lasting impact seeing as the number remained strong at 61% in 2022 (SFA & Accenture, 2022). Fintech firms have a derived demand for labor, especially in the technology and business development areas as seen in *fig. 5*. A rightward shift in demand for digital processes in the fintech market translates to a derived demand to increase headcount in its labor market.

6.2 Non-monetary Employee Benefits

From what has been deduced in *section 5*, the appeal of the fintech sector is not its wages or job security. Hence, the government, data analysts and industry experts recommend firms to differentiate themselves by focusing on non-wage factors (Violino, 2023).

The changes in work culture and attitude in COVID-19 includes individuals looking beyond high compensations, balancing it with a desire for work-life balance. For example, 96% of employees feel that work flexibility is important when seeking a new job (Robinson, 2022). Fortunately, flexible work arrangements (telecommuting and flexible leave) are offered by approximately 86% of fintech firms (SFA & Accenture, 2021). Thus, this approach was favorable to fintech as it overlooks their shortcomings of offering a relatively lower salary.

The 2022 Fintech Talent Report corroborates this argument: fintech workers expressed that Employee Value Propositions (EVPs) were the main reason employees decided to join the fintech workforce. EVPs in this context mainly focus on opportunities that foster continuous growth, talent mobility, meaningful work that aligns with individual aspirations, a collaborative work environment and tools that drive workplace engagement, health and safety (SFA & Accenture, 2022).

6.3 Upskilling the Workforce

Upskilling is essential for two reasons.

Firstly, despite EVPs being the main incentive for candidates to join fintech firms, it is also the top reason employees decide to leave their jobs, resulting in an increasingly high turnover rate. In 2020, the average turnover in the finance sector is 6.1%; in 2022, approximately 42% of fintech firms experienced a turnover rate of 10% to 20%, the second highest turnover rate globally (Asian Banking & Finance, 2022; Chong, 2022; Ministry of Trade and Industry, 2022b). This is detrimental as high turnover rates mean firms lose productivity from additional recruitment costs, low morale of employees (Staw, 1980).

Secondly, the average shelf-life of tech skills is less than five years and for more specialized fields, it can be as low as two and a half years (Malik, 2020). With rapid

advancements of technology, AI and other machinery can carry out sophisticated tasks such as research, coding and writing that used to be considered unsubstitutable by capital (Tamayo, Doumi, Goel, Kovács-Ondrejkovic, & Sadun, 2023). Upskilling is therefore a priority for professionals to renew their skills as well as diversifying them.

The private sector recognizes the importance of training, especially for technical skills, seeing as skills such as strategy planning and project management are by far the dominating training area. However, the amount spent on training remains insufficient (SFA & Accenture, 2022). In 2021, the global average spent on training and development per employee was $1,270 (~SDG1728) , yet over 70% of fintech firms spend less than SDG1000 per employee in both 2021 and 2022 (SFA & Accenture, 2022; Statista Research Department, 2023b). This provides an explanation for employees leaving firms because of the lack of career development and enhancement.

The government provided grants like the IBF (Institute of Banking and Finance) funding to encourage firms to upskill their employees. However, this, like various other grants, achieved marginal impact as 87% of producers did not leverage it, mainly due to the lack of awareness (SFA & Accenture, 2022).

6.4 Reskilling the Workforce

Recently, a new argument has been proposed by many experts (Tamayo, Doumi, Goel, Kovács-Ondrejkovic, & Sadun, 2023). As the entire technology industry is struggling with the labor shortage, upskilling is insufficient because it only improves the quality of the existing labor without increasing the quantity of tech talent. Thus, many believe that the better approach would be to reskill the workforce.

The government offers robust career conversion programs and schemes that encourage financial institutions to hire “workers from other sectors and place them in talent

development programs" by doubling their salary support (Singapore Fintech Association, 2020; Workforce Singapore, 2023).

However, as this approach has not been sufficiently appreciated by the private sector seeing that firms only seek tech talent with experience (at least one year of work experience) and insufficient work experience is also one of the main reasons for firms to reject a candidate even if they have completed a training course provided by the government (SFA, Banking and Financial Services Union, & PwC, 2020). This suggests that the curriculum for such programs may lack sufficient rigor or sophistication to equip participants with skills that meet the needs of the industry or provide a seamless mid-career switch.

7 Alternatives

7.1 Artificial Intelligence and Other Capital Substitution to Increase Productivity

In 2020, digitizing business processes was the most common approach to increase productivity in fintech at 57.6%, compared to retraining staff which 24.2% of businesses adopted (Singapore Fintech Association, Banking and Financial Services Union, & PwC, 2020).

But in 2022, labor productivity in the finance sector declined by 2.7% real value added per actual hour worked (Ministry of Trade and Industry, 2022a). A likely explanation is the lack of attention on health and well-being of employees, where 99% of finance professionals in Singapore experience some level of burnout (Whitmire, 2023).

In May 2023, MAS and Google signed a memorandum stating that Google will facilitate in improving the AI skills of MAS technologists to execute the adoption of AI in financial services in creating generative AI applications for both external and internal use. The Singaporean government has already invested SDG1.87B in this market. While this is a relatively new initiative so little information can be drawn on its effectiveness, it has been

observed in a few other sectors that adopting AI made 81% of employees designate their work to AI to lessen their workload by facilitating "process automation, advanced analytics and managed services" all of which can improve productivity (Bhatt, 2023).

Moreover, they significantly complement the current on-the-job training model, leveraged by many fintech firms as it can act as a virtual mentor (Violino, 2023). As seen in the field of cybersecurity that faced severe talent shortages (Hirdaramani, 2022; Violino, 2023), using generative AI allowed senior members to focus on their own tasks, preventing any hindrance on labor productivity. The overall effect of digitalization can facilitate potential growth, as observed in *fig. 6* as PPC_1 shifts outwards to PPC_2, and by extension, prompt actual growth, shifting actual output from point A to B.

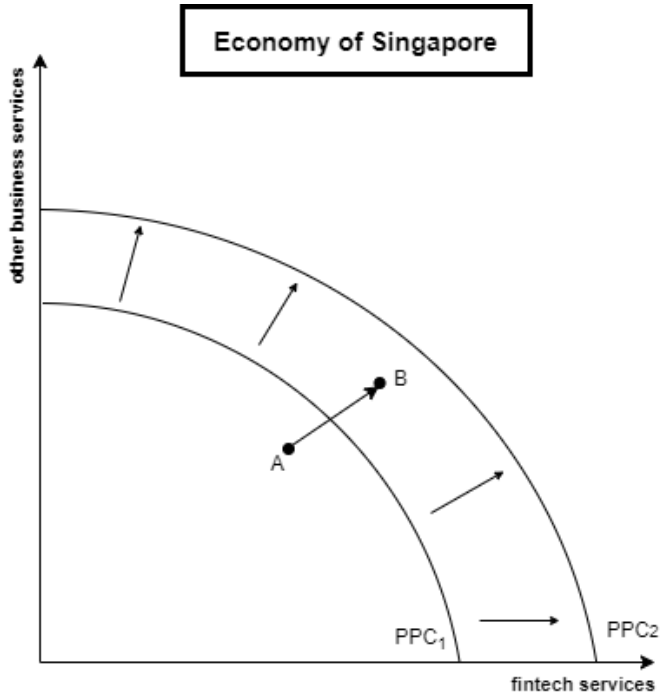


Figure 6. Effect of Digitalization and AI on the PPC of Singapore

7.2 Foreign Workers

As of 2022, 34% of the country's workforce is composed of foreign workers (Ministry of Manpower, 2022). However, the need for foreign workers is quite a controversial topic within the country. In 2020, the skills gap between foreign and local tech workers widened to 7%,

compared to previous years (Seow, 2021). This illustrates the need for foreign talent: their higher quality in skill could theoretically bring higher productivity. It can therefore be assumed that less onboarding training is needed as they can apply their existing skills.

But due to the pandemic and ongoing protests of the middle and lower-class, the government has restricted foreign talent from joining Singapore's workforce. The government has a soft and indecisive stance seeing as they have tightened immigration policies and increased thresholds for work permits (Aravindan & Geddie, 2020), yet they also wish to avoid the pretense that their economy is not open to international interactions.

However, the government does not view this as a sustainable solution because as countries progress, and opportunities increase, more benefits will be required to incentivize foreign workers to move to Singapore to work (Hanhonn, 2022). Hence, the government has continued to heavily train its local workforce, generously giving resources and encouraging innovation to aptly equip them with skills needed to face international competitors (Subhani, 2022). There is also a shift in hiring more local employees (Singapore Fintech Association, Banking and Financial Services Union, & PwC, 2020).

Conversely, many working in the fintech industry have expressed the need to make it easier to obtain work permits for foreign employees to close the talent gap. Indeed, in 2022, as border restrictions were lifted, job vacancies declined from 35% to 27% in a year (Ong, 2023).

The finding that can be drawn from this is that while foreign workers are an attractive solution for the private sector, both producers and the government recognize its unsustainability which explains the emphasis on local labor.

8 Conclusion

As technological skills in fintech continue to be in high demand, both wage and non-wage factors must be considered to increase the supply of labor. However, through the evaluation of the wage inelasticity of labor supply along with the stance of fintech firms, it is evident that non-wage factors including EVPs and career development unlock greater incentives for candidates to join the industry.

The public sector spreads their resources to cover both wage and non-wage factors, in the form of subsidies and grants. They also support the private sector to increase productivity by investing in key technology and facilitating training. However, the impact brought by many of their initiatives fall short due to the private sector being unaware of such initiatives. Additionally, in areas such as inflated wages, tightening of work permits and increased training programs, their initiatives contradict the needs of the private sector.

The private sector concentrates its investments into two key areas: career development to attract candidates and retain existing employees and therefore increasing supply, and increasing productivity through digitalization of business procedures and training to decelerate the rise in demand. Unfortunately, despite their best efforts, the funds they allocate are often insufficient to achieve these goals. Nonetheless, firms are quick to adapt to changes in mindset and technology, especially when aligning themselves with government plans, such as incorporating a hybrid work arrangement or the adoption of AI which has been shown to give the industry an upper hand.

These findings align with those witnessed in Germany's VET system. By evaluating the actions taken by both the public sector to correct market failure and the private sector to correct government failure in the labor market, this essay unveils the complementary relationships between the two, in achieving their individual goals regarding human capital.

9 References

Amt, A. (n.d.). The German Vocational Training System: An Overview. Retrieved from www.germany.info website: https://www.germany.info/us-en/welcome/wirtschaft/03-Wirtschaft/-/1048296

Aravindan, A., & Geddie, J. (2020, September 2). Singapore PM warns against turn inwards as anxiety over foreigners grows. *Reuters*. Retrieved from https://www.reuters.com/article/us-singapore-immigration-idUSKBN25T1KA

Asian Banking & Finance. (2022, May 19). Singapore fintech talents record 2nd highest turnover rate globally: study. Retrieved November 12, 2023, from Asian Banking & Finance website: https://asianbankingandfinance.net/news/singapore-fintech-talents-record-2nd-highest-turnover-rate-globally-study

Bayo-Moriones, A., Galdón-Sánchez, J. E., & Güell, M. (2004). *Is Seniority-based Pay Used as a Motivation Device?*

Bhatt, S. (2023, July 18). How will AI Take Forward the FinTech Industry in Singapore. Retrieved November 12, 2023, from Tntra website: https://www.tntra.io/blog/how-will-ai-take-forward-the-fintech-industry-in-singapore/#:~:text=The%20MAS%20and%20Google%20have

Bouteiller, F., & Carlei, S. (2020, March 16). Singapore - Covid-19 - Immigration update. Retrieved November 12, 2023, from Deloitte Société d'Avocats website: https://blog.avocats.deloitte.fr/singapore-covid-19-immigration-update

Chandra, K. (2018). 56% employers in Singapore report difficulty filling positions, highest reported since 2008. Retrieved from Manpower Group website: https://www.manpower.com.sg/blog/2018/07/56-percent-employers-in-singapore-report-difficulty-filli%20ng-positions-highest-reported-since-2008?source=google.com

Chanjaroen, C. (2022, September 15). Singapore to Create Up to 20,000 Finance Jobs Over Five Years. *Bloomberg*. Retrieved from https://www.bloomberg.com/news/articles/2022-09-15/singapore-to-create-up-to-20-000-finance-jobs-over-five-years

Cheng, I. (2023, July). Singapore birth rates hit record low in 2022, along with highest number of yearly deaths since 1960. The Strait Times. Retrieved 28 August 2023, from https://www.straitstimes.com/singapore/singapore-s-birth-rates-hit-record-low-in-2022-along-with-most-yearly-deaths-since-1960#:~:text=SINGAPORE%20%2D%20Singapore's%20birth%20rate%20hit,2021%20to%2035%2C605%20in%202022.

Chew, H. M. (2021a, May). Finance sector projected to create 6,500 jobs in 2021, building on last year's expansion. Retrieved November 12, 2023, from Channel News Asia website: https://www.channelnewsasia.com/business/finance-sector-create-6500-jobs-2021-tech-consumer-banking-1388041

Chew, H. M. (2021b, May). Financial tech workers in demand in Singapore, getting multiple job offers and pay increments. Retrieved from Channel News Asia website:

https://www.channelnewsasia.com/business/finance-tech-jobs-singapore-salary-fintech-1398826

Chiang, S. (2023, March 7). Singapore's tech salaries mostly jumped in 2022, new report shows. Retrieved from CNBC website: https://www.cnbc.com/2023/03/07/singapores-tech-salaries-mostly-jumped-in-2022-new-report-shows.html

Chong, C. (2022, November). Attrition rates a concern to fintech firms facing "opportunity paradox." Retrieved November 12, 2023, from The Business Times website: https://www.businesstimes.com.sg/startups-tech/startups/mobile-spotlight/attrition-rates-concern-fintech-firms-facing-opportunity#:~:text=A%20survey%20conducted%20by%20the

Deloitte. (2015). The youth employment opportunity. Retrieved from Deloitte Turkey website: https://www2.deloitte.com/tr/en/pages/about-deloitte/articles/youth-employment-opportunity.html

Department of Statistics Singapore. (2023, March). Singapore Economy. Retrieved from Department of Statistics Singapore website: https://www.singstat.gov.sg/modules/infographics/economy

Digital News Asia. (2023, January 13). Tech jobs outpace Singapore's overall job growth in 2022, demand for tech talent remains: Seek. Retrieved November 11, 2023, from Digital News Asia website: https://www.digitalnewsasia.com/business/tech-jobs-outpace-singapores-overall-job-growth-2022-demand-tech-talent-remains-seek

ExplainThis. (2023, August 31). 在新加坡當工程師薪水有多少？(2023) | ExplainThis.

Retrieved November 12, 2023, from ExplainThis website: https://www.explainthis.io/zh-hant/career/software-engineer-salary-in-singapore

Federal Institute for Vocational Education and Training. (n.d.). Federal Institute for Vocational Education and Training (BIBB) - Germany. Retrieved November 11, 2023, from BIBB website: https://www.bibb.de/en/77212.php

Fintech News Singapore. (2020, August 13). MAS Commits S$250 Million to Drive Innovation and Build Fintech Talent Pipeline. Retrieved November 11, 2023, from Fintech News (Singapore) website: https://fintechnews.sg/42797/fintech/mas-commits-s250-million-to-drive-innovation-and-build-fintech-talent-pipeline/

Fu, J., & Mishra, M. (2021). Fintech in the time of COVID-19: Technological adoption during crises. *Journal of Financial Intermediation*, *50*(100945), 100945. https://doi.org/10.1016/j.jfi.2021.100945

Fürstenau, B., Pilz, M., & Gonon, P. (2014). The Dual System of Vocational Education and Training in Germany – What Can Be Learnt About Education for (Other) Professions. *International Handbook of Research in Professional and Practice-Based Learning*, 427–460. https://doi.org/10.1007/978-94-017-8902-8_16

GOVET. (n.d.). Govet / Cooperation of government and industry. Retrieved from BIBB - Cooperation of government and industry website: https://www.govet.international/en/13711.php

Haasler, S. R. (2020). The German system of vocational education and training: challenges of gender, academisation and the integration of low-achieving youth. *Transfer: European Review of Labour and Research*, *26*(1), 57–71. https://doi.org/10.1177/1024258919898115

Hanhonn, I. T. (2022, November 16). Zaqy Mohamad: Singapore Must Reduce Reliance on Foreign Manpower. Retrieved November 12, 2023, from National Trades Union Congress website: https://www.ntuc.org.sg/uportal/news/Zaqy-Mohamad-Singapore-Must-Reduce-Reliance-on-Foreign-Manpower/

Harry, C. (2020, July 1). FinTech Employee Retention. Retrieved from Storm2 website: https://storm2.com/resources/recruitment/fintech-employee-retention/

Harvey, É. (1993). Youth Unemployment and Policies towards Unemployed Youth, 1918–1933. In *Oxford University Press eBooks* (pp. 103–151). Oxford Historical Monographs . https://doi.org/10.1093/acprof:oso/9780198204145.003.0004

Hirdaramani, Y. (2022, October 17). Cybersecurity key to the future of Singapore's financial services. Retrieved from GovInsider website: https://govinsider.asia/intl-en/article/cybersecurity-key-to-the-future-of-singapores-financial-services

Horton, C. (n.d.). Tech talent shortage – why it's time to get outside your comfort zone. Retrieved from ERP Today website: https://erp.today/tech-talent-shortage-why-its-time-to-get-outside-your-comfort-zone/

Indeed Editorial Team. (2023, June). 22 Most In-Demand Jobs in Singapore (Including Salaries). Retrieved from Indeed website: https://sg.indeed.com/career-advice/finding-a-job/most-in-demand-jobs-in-singapore

IndSights Research. (2023, October 27). FinTech in Singapore. Retrieved from Indsights website: https://www.indsights.sg/industry-perspective/fintech-in-singapore/#:~:text=FinTech%20in%20Singapore%20will%20promote

Inland Revenue Authority Of Singapore. (2023, September). Jobs Support Scheme (JSS). Retrieved from Inland Revenue Authority Of Singapore website: https://www.iras.gov.sg/schemes/disbursement-schemes/jobs-support-scheme-(jss)

INSIGHT. (2022, June 7). Singapore Tech Salaries Guide (2022): Are You Underpaid? Retrieved from Nodeflair website: https://nodeflair.com/blog/singapore-tech-salaries-guide

Insight. (2023, January 17). Top 10 Fintech Companies to Work for in 2023 (based on REAL Salary and Reviews). Retrieved from Nodeflair website: https://nodeflair.com/blog/top-10-fintech-companies-to-work-for-in-2023-based-on-real-salary-and-reviews

Jacoby, T. (2014, October 16). Why Germany Is So Much Better at Training Its Workers. Retrieved from The Atlantic website: https://www.theatlantic.com/business/archive/2014/10/why-germany-is-so-much-better-at-training-its-workers/381550/

Keloth, R. (2022, April). Singapore is changing its work visa rules for foreign employees. Retrieved from Intertrust Group website: https://www.intertrustgroup.com/insights/singapore-is-changing-its-work-visa-rules-for-foreign-employees/

Kumar, S., & Siddique, S. (2010, March). Ths Singapore Success Story: Public-Private Alliance for Investment Attraction, Innovation and Export Development. Retrieved from Singapore Case Formato Final website: https://repositorio.cepal.org/server/api/core/bitstreams/66fc9d92-7417-4167-9b69-8f322d58d31b/content

lanturn. (2020, August 3). 3 Main Challenges of Getting a Singapore Work Visa Approved | Lanturn Learning. Retrieved from Lanturn website: https://lanturn.com/sg/blog/3-main-challenges-of-getting-a-singapore-work-visa-approved/

Lee, H. L. (2023, April). PM Lee Hsien Loong at the 'Singapore Ageing: Issues and Challenges Ahead' Book Launch. Prime Minister's Office Singapore. Retrieved 28 August 2023, from https://www.pmo.gov.sg/Newsroom/PM-Lee-Hsien-Loong-at-the-Singapore-Ageing-Issues-and-Challenges-Ahead-Book-Launch#:~:text=Our%20population%20is%20not%20just,in%204%20Singaporeans%20over%2065.

Lim, C. L. (2022, August 29). Banking and Financial Services Job Market and Salary Outlook in Singapore 2022 | Randstad SingaporeRandstad Singapore. Retrieved November 11, 2023, from Randstad website: https://www.randstad.com.sg/hr-trends/workforce-trends/banking-financial-services-job-market-salary-outlook-singapore-2022/

Lüdtke, L. (2018, December 11). Vocational lessons from Germany and France. Retrieved from GIS Reports website: https://www.gisreportsonline.com/r/germany-vocational-education-system/

Macrotrends. (2023.) Singapore Birth Rate 1950-2023. Retrieved from Macrotrends website: https://www.macrotrends.net/countries/SGP/singapore/birth-rate#:~:text=The%20current%20birth%20rate%20for,a%200.96%25%20decline%20from%202020.

Macrotrends. (2023.) Singapore Population Growth Rate 1950-2023. Retrieved from Macrotrends website: https://www.macrotrends.net/countries/SGP/singapore/population-growth-rate

Malik, S. (2020, December 7). Skills Transformation For The 2021 Workplace. Retrieved from IBM Training and Skills Blog website: https://www.ibm.com/blogs/ibm-training/skills-transformation-2021-workplace/

ManpowerGroup. (2022). What Workers Want 2019 Talent Shortage Study. Retrieved from Manpower Group website: https://go.manpowergroup.com/talent-shortage

ManpowerGroup. (2023). Global Talent Shortage 2023 Global Average. In *Manpower Group*. Retrieved from https://go.manpowergroup.com/hubfs/MPG_TS_2023_Infographic_FINAL.pdf

Ministry of Education. (2020). *Education Statistics Digest 2020*. Ministry of Education.

Ministry of Education. (2021). *Education Statistics Digest 2021*. Ministry of Education.

Ministry of Education. (2022). *Education Statistics Digest 2022*. Ministry of Education.

Ministry of Finance. (2022, December 6). Economy and Labour Market. Retrieved November 12, 2023, from Singapore Public Sector Outcomes Review website: https://www.mof.gov.sg/singapore-public-sector-outcomes-review/businesses/strong-and-resilient-economy/economy-and-labour-market#:~:text=In%202021%2C%20our%20economy%20expanded

Ministry of Manpower. (2022, December). Foreign workforce numbers. Retrieved from Ministry of Manpower Singapore website: https://www.mom.gov.sg/documents-and-publications/foreign-workforce-numbers

Ministry of Trade and Industry. (2022a). LABOUR MARKET AND PRODUCTIVITY. In *Economic Survey of Singapore 2022*. Retrieved from https://www.mti.gov.sg/-/media/MTI/Resources/Economic-Survey-of-Singapore/2022/Economic-Survey-of-Singapore-2022/Ch2_AES2022.pdf

Ministry of Trade and Industry. (2022b, February). Economic Survey of Singapore 2021. Retrieved from Ministry of Trade and Industry website: https://www.mti.gov.sg/-/media/MTI/Resources/Economic-Survey-of-Singapore/2021/Economic-Survey-of-Singapore-2021/FullReport_AES2021.pdf

Monetary Authority of Singapore. (2020, April). MAS Launches S$125 Million Package for Financial Institutions and FinTech Firms to Strengthen Long-Term Capabilities. Retrieved from Monetary Authority of Singapore website: https://www.mas.gov.sg/news/media-releases/2020/mas-launches-package-for-fis-and-fintech-firms-to-strengthen-long-term-capabilities#:~:text=2%20The%20support%20package%2C%20funded,to%20digital%20platforms%20and%20tools.

Monetary Authority of Singapore. (2023, April). COVID-19 Support Package for FinTechs. Retrieved from Monetary Authority of Singapore website: https://www.mas.gov.sg/development/fintech/covid-19-support-package-for-fintechs

O'Neill, A. (2023a, August 17). Global youth unemployment rate until 2019. Retrieved from Statista website: https://www.statista.com/statistics/269636/global-youth-unemployment-rate/

O'Neill, A. (2023b, October 4). Germany - youth unemployment rate 1998-2018 | Statista. Retrieved from Statista website: https://www.statista.com/statistics/812019/youth-unemployment-rate-in-germany/

Ong, J. (2023, March 30). Job vacancies decline as borders reopen, tech talent still in high demand across many sectors: MOM. Retrieved November 12, 2023, from TODAY website: https://www.todayonline.com/singapore/job-vacancies-decline-tech-talents-demand-2140736

Ozturk, I. (2001). The role of education in economic development: a theoretical perspective. *Journal of Rural Development and Administration*, *XXXIII*(1), 39–47. Retrieved from https://mpra.ub.uni-muenchen.de/id/eprint/9023

Payments Industry Intelligence. (2023, July 14). How AI is changing fintech jobs. Retrieved from Payments Cards and Mobile website: https://www.paymentscardsandmobile.com/how-ai-is-changing-fintech-jobs/

Pinnow, S. (2019, December 19). The German Dual VET System and Approaches to enhance Employer Engagement. Retrieved November 11, 2023, from TVET@Asia website: https://tvet-online.asia/13/pinnow/#:~:text=The%20origins%20of%20the%20German

Robinson, B. (2022, May 1). Work Flexibility Is The Most Important Leadership Skill, According To Research. Retrieved November 11, 2023, from Forbes website: https://www.forbes.com/sites/bryanrobinson/2022/05/01/work-flexibility-is-the-most-important-leadership-skill-according-to-research/?sh=647f82ba4a04

See, S. (2025, March 15). Return of foreign workers sends Singapore's total employment growth to record level in 2022. Retrieved from The Business Times website: https://www.businesstimes.com.sg/singapore/return-foreign-workers-sends-singapores-total-employment-growth-record-level-2022

Seow, J. (2021, May 16). What is stopping more Singaporeans from taking up tech jobs? *The Straits Times*. Retrieved from https://www.straitstimes.com/business/invest/embarking-on-a-tech-career

SG United. (2020, April 8). COVID-19 FinTech Care Package. Retrieved November 12, 2023, from Monetary Authority of Singapore website: https://www.mas.gov.sg/-/media/mas/fintech/ftig-covid-icons/covid-package-infographic.jpg

Singapore Business Review. (2023a, September 26). Singapore named Asia's top fintech hub, fourth globally. Retrieved November 1, 2023, from Singapore Business Review website: https://sbr.com.sg/financial-services/news/singapore-named-asias-top-fintech-hub-fourth-globally

Singapore Business Review. (2023b, September 29). Singapore's fintech to contribute 3.6% in 2027. Retrieved from Singapore Business Review website: https://sbr.com.sg/financial-services/news/singapores-fintech-contribute-36-in-2027

Singapore Fintech Association. (2020, September). Grants Portal. Retrieved November 12, 2023, from Singapore Fintech Association website: https://singaporefintech.org/grants-portal/

Singapore Fintech Association, & Accenture. (2021). Fintech talent report 2021. Retrieved from Singapore Fintech Association website: https://singaporefintech.org/wp-content/uploads/2023/05/SFA-Accenture-FinTech-Talent-Report-2021.pdf

Singapore Fintech Association, & Accenture. (2022). Fintech talent report 2022. In *Singapore Fintech Association*. Retrieved from https://nexusfrontier.tech/wp-content/uploads/2022/11/SFA-Accenture-FinTech-Talent-Report-2022-FINAL.pdf

Singapore Fintech Association, Banking and Financial Services Union, & PwC. (2020). FinTech talent survey 2020: Seizing opportunities for growth. In *Singapore Fintech Association*. Retrieved from https://www.pwc.com/sg/en/financial-services/assets/seizing-opportunities-for-growth-fintech-talent-survey.pdf

Singapore Fintech Association, & Oliver Wyman. (2020). Singapore Fintech Landscape 2020 and Beyond. In *Singapore Fintech Association*. Retrieved from https://www.oliverwyman.com/content/dam/oliver-wyman/v2/publications/2020/dec/singapore-fintech-landscape-2020-and-beyond.pdf

Singapore Fintech Association, & PwC. (2019). Fintech talent survey 2019. Retrieved from Singapore Fintech Association website: https://www.pwc.com/sg/en/publications/assets/fintech-talent-survey-2019.pdf

Singapore's poor productivity performance. (2020, November 20). Retrieved from Academia | SG website: https://www.academia.sg/academic-views/singapores-poor-productivity-performance/

Specs, A.-C. (2018, June 12). Could Germany's Vocational Education Training System Be a Model for the U.S.? Retrieved from World Education News + Reviews website: https://wenr.wes.org/2018/06/could-germanys-vocational-education-and-training-system-be-a-model-for-the-u-s

Statista Research Department. (2023a, February 20). Singapore: fintech investments value 2021. Retrieved from Statista website: https://www.statista.com/statistics/1010180/singapore-fintech-investments-value/

Statista Research Department. (2023b, April). Average spend on workplace training per employee worldwide from 2008 to 2021. Retrieved from Statista website: https://www.statista.com/statistics/738519/workplace-training-spending-per-employee/#:~:text=In%202020%2C%20employers%20worldwide%20spent

Statista Research Department. (2023c, May). Singapore: GDP growth by industry 2019. Retrieved from Statista website: https://www.statista.com/statistics/862725/singapore-gdp-growth-by-industry/

Statista Research Department. (2023d, September 4). Singapore: job vacancies in financial services 2022. Retrieved November 12, 2023, from Statista website: https://www.statista.com/statistics/976344/job-vacancy-financial-services-singapore/

Staw, B. M. (1980). The Consequences of Turnover. *Journal of Occupational Behaviour*, *1*(4), 253–273. Retrieved from http://www.jstor.org/stable/3000143

Subhani, O. (2022, December 5). Staying open to foreign talent helps ensure long-term economic growth: Chee Hong Tat | The Straits Times. Retrieved from The Straits Times website: https://www.straitstimes.com/business/singapore-needs-foreign-talent-to-ensure-long-term-economic-growth-to-everyone-s-benefit

Tamayo, J., Doumi, L., Goel, S., Kovács-Ondrejkovic, O., & Sadun, R. (2023, September 1). Reskilling in the Age of AI. Retrieved November 12, 2023, from Harvard Business Review website: https://hbr.org/2023/09/reskilling-in-the-age-of-ai#:~:text=And%20Amazon%2C%20through%20its%20Machine

TechNode Global. (2023, September 21). UnaFinancial: share of fintech in the GDP of Singapore will reach 3.56 percent in 2027. Retrieved November 12, 2023, from TNGlobal website: https://technode.global/2023/09/21/unafinancial-share-of-fintech-in-the-gdp-of-singapore-will-reach-3-56-percent-in-2027/

Tenity. (2023, February 20). The Power of the Singapore Fintech Ecosystem - Tenity. Retrieved November 12, 2023, from Tenity website: https://www.tenity.com/blog/the-power-of-the-singapore-fintech-ecosystem#:~:text=Strong%20financial%20industry

The Strait Times. (2023). Singapore salary guide 2023. Retrieved from The Strait Times website: https://www.straitstimes.com/multimedia/graphics/2023/07/salary-guide-2023/index.html?shell

The Talent Issue: Local and Foreign Employees Needed for Organisational Success. (2021, February 24). Retrieved from MyCareersFuture website: https://content.mycareersfuture.gov.sg/talent-issue-local-foreign-employees-needed-organisational-success/#:~:text=In%20Singapore%2C%20foreign%20labour%20helps

Tong, G. C. (2023, January). Singapore's 15 fastest-growing jobs for 2023, according to LinkedIn. Retrieved from CNBC website: https://www.cnbc.com/2023/01/22/singapores-15-fastest-growing-jobs-for-2023-according-to-linkedin.html

Violino, B. (2023, September 27). How AI and better pay can address the ongoing cyber talent shortage. Retrieved November 11, 2023, from CNBC website: https://www.cnbc.com/2023/09/27/how-ai-and-better-pay-can-address-the-ongoing-cyber-talent-shortage.html

Whitmire, M. (2023, February). How Fintech Is Poised for Transformation | Built In. Retrieved November 12, 2023, from builtin website: https://builtin.com/fintech/fintech-industry-trends

Workforce Singapore. (2023, February 10). Workforce Singapore launches first Career Conversion Programme in FinTech industry with Singapore FinTech Association as programme partner. Retrieved from Workforce Singapore website: https://www.wsg.gov.sg/home/media-room/media-releases/workforce-singapore-launches-first-career-conversion-programme-in-fintech-industry-with-singapore-fintech-association-as-programme-partner

Worldometer. (2023, July 16). Singapore Population (2022) - Worldometers. Retrieved from Worldometers.info website: https://www.worldometers.info/world-population/singapore-population/

Yahya, F. B. (2022, December 2). Singapore's tech industry starved for global talent. Retrieved from Asia Times website: https://asiatimes.com/2022/12/singapores-tech-industry-starved-for-global-talent/

Yi, T. H. (2023, March 30). Tech talent still highly sought after in S'pore, MOM job vacancy report shows | The Straits Times. Retrieved from The Straits Times website: https://www.straitstimes.com/singapore/jobs/tech-talent-still-highly-sought-after-in-s-pore-mom-job-vacancy-report-shows

10 Appendices

Appendix A

		2021	2022
Demand	Vacant tech jobs	19,000	20,350

	New jobs in fintech	1,250	3,000
	Proportion of foreign labor in fintech	35.00%	33.33%
	Proportion of vacant fintech jobs among total vacancies	5.90%	14.74%
	Domestic tech labor demand	8,850	8,983
	Domestic tech labor demand for fintech	1,250	3,000
Supply	IT graduates from ITE, polytechnic or university	6,925	7,087
	Estimated number of graduates moving into fintech	408	1,045
Total shortage of domestic tech labor		1,925	1,896
Shotage of domestic tech labor in fintech		**842**	**1,955**

NOTE: Data for 2021 were from sources collected in 2020 and data from 2022 were from sources collected in 2021